

A DIRECTIONAL DEFAULT, NOT A DIRECTIVE EFFECT: BEHAVIORAL MEASUREMENT OF POLITICAL FRAM- ING IN FRONTIER LANGUAGE MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

Neutrality directives—system-prompt instructions such as “*do not editorialize*”—are widely proposed as a remedy for political bias in LLMs, and LLM judges are the default instrument for auditing them. This paper subjects both to a rule the judge literature lacks: no instrument’s output counts until it passes a pre-committed, cross-family reliability gate. The gates exposed a measurement hazard: two of three plausible instruments failed calibration systematically, and ungated judging had already produced a publishable artifact—an apparent 55–64% rate of framing absorption that fell to 0–13% (per-judge range) under an evidence-anchored rubric. On the surviving instrument, a pre-registered two-stage study of two frontier models on 200 outlet-anchored articles dissociates decisions from rationales. The directive left decisions statistically equivalent (detection counts within ± 1.0 per article, TOST; lean labels stable, $\kappa = 0.90$), while rationales carry a directional default no tested intervention causes or removes: framing in right-leaning sources is substituted toward the center at $1.9\times$ the mirror rate (11.0% vs. 5.8%, $p = .0014$; permutation $p = 10^{-4}$), present at baseline, confined to one model (+10.1pp vs. +0.1pp, CI $[-3.0, +3.2]$), and corroborated by a judge-free lexicon instrument (pooled +5.8pp, $p = .03$). The instrument’s measured limits (item-level κ 0.606 at the gate, 0.255 at scale; sign agreement 93.5%; the asymmetry survives restriction to both-judge-agreed items, $p = .005$) grade the claim, pending human arbitration. The gated audit re-runs per model generation for $\sim \$150$ in marginal API cost, and each finding maps to a design rule for news-facing AI products.

1 INTRODUCTION

When a language model is asked to analyze a politically charged news article, it produces two things: a *decision* (an answer—which spans are biased, what the article’s lean is) and a *rationale* (a stated explanation of that answer). Much of the governance of political behavior in deployed models operates on the assumption that these move together: add a neutrality directive to the system prompt, observe better-sounding explanations, and conclude the model has become more even-handed. Each step of that pattern is documented practice, not caricature: prompt-level neutrality directives are the recurrent mitigation in the research literature (Ganguli et al., 2023; Liu et al., 2025; Yang et al., 2025; Fisher et al., 2025) and ship verbatim in production system prompts (Anthropic, 2025); reading model-verbalized reasoning is itself institutionalizing as a governance instrument (Korbak et al., 2025; Baker et al., 2025); and under the 2025 U.S. procurement rules a vendor may demonstrate “ideological neutrality” by disclosing the system prompt itself (Executive Office of the President, 2025; Office of Management and Budget, 2025). This paper tests the underlying assumption directly, and pre-registers the prediction that it is false in a specific, directional way.

Our central design principle is a dissociation test. If a directive genuinely alters processing, its effects should appear in the decisions themselves. If instead the directive is absorbed as a *presentation instruction*, decisions will be unchanged while rationales are rewritten to *perform* compliance. And there is a third possibility—the one our data ultimately support: the directive moves neither channel, and the political signature is a resting-state property of the rationale channel alone, present before

any instruction and removed by none we test. The distinction matters for AI safety practice well beyond the political domain: it determines whether behavioral audits and self-report audits of the same system can be expected to agree. A model whose explanations respond to instructions that its decisions ignore will pass every audit that reads its explanations and fail every audit that measures its behavior—or, more dangerously, the reverse. Nor is the setting hypothetical: the nearest deployment surface is the growing class of news-facing AI products—bias checkers, balanced-news summarizers, media-literacy tools built on expert outlet ratings—which surface exactly these two channels to readers: decisions (flagged spans, lean labels) and explanatory prose. Whether the two channels can diverge, and in which direction, is for that product class a correctness property, not an abstraction; §5 translates each finding into a design rule for it.

Testing this required confronting a second problem, which became a co-equal contribution of the paper. At the scale of thousands of judgments, the only affordable instruments are LLM judges, and a common practice is to write a plausible rubric, spot-check a few outputs, and trust the resulting numbers. We instead adopted the standard that psychometrics applies to any new instrument: **demonstrate reliability before collecting confirmatory data, under acceptance criteria committed in advance**. Concretely, every instrument was scored independently by two frontier judges from different model families (cross-family pairing, to break same-family self-preference circularity; Panickssery et al., 2024), on held-out pilot data excluded from all confirmatory analysis, against a pre-committed gate (Cohen’s $\kappa \geq 0.40$ plus non-degeneracy conditions), with a pre-registered fallback and a one-shot rule forbidding iterative re-anchoring until the gate passes.

The gates did real work. Of three instruments that all looked reasonable on paper—and whose pilot outputs all *looked* sensible to inspection—two failed calibration in ways that would have silently corrupted the study (§4.1). Most consequentially, an ungated instrument had already minted the pilot’s most publishable number—an apparent 55–64% rate at which models absorb loaded source framing into their own explanatory voice. Forensic audit attributed most of it to judge error (with a quantified corpus-error component; §4.1): under a corrected, evidence-anchored rubric the rate collapsed to 0–13% depending on judge. This trajectory is itself the paper’s first result: plausible-looking LLM-judge instruments can fail cross-family calibration—in this study, two of three did—the failures are systematic rather than noisy, and ungated instruments convert judge artifacts into publishable effects.

Contributions.

1. **A directional default, not a directive effect.** The neutrality directive changed neither channel: decisions remained statistically equivalent across the contrast, while rationales carried a directional asymmetry—framing in right-leaning sources re-coded toward the center at $1.9\times$ the mirror rate—that predates the directive and survives every intervention we tested. All three surviving pre-registered tests confirmed; both judge families independently significant; corroborated by a judge-free lexicon instrument. The effect is confined to one of the two target models (Claude Sonnet 4.5, +10.1pp; GPT-4.1, null), and the registered directive→rationale-style tests, demoted with their failed instruments, remain exploratory.
2. **Instruments must earn trust before their output counts.** Every LLM judge faced a pre-committed, cross-family reliability gate: two of three plausible instruments failed—systematically, not noisily—and the gates caught a publishable artifact before it became the headline finding (an apparent 55–64% framing-absorption rate that fell to 0–13% under an evidence-anchored rubric). The failure modes, and the design features that saved the surviving instrument, are characterized for reuse (Figure 1).
3. **A refreshable audit, not a one-off study.** A deterministic corpus recipe over expert-rated news, gated instruments, and analysis code, re-runnable on any future model generation for ~\$150 in marginal API cost (validation layers priced separately). The same package is a pre-deployment qualification test for news-facing AI features (§5)—and the behavioral-verification layer that disclosure-based neutrality compliance (Office of Management and Budget, 2025) currently accepts on trust.

We emphasize the asymmetry between our findings and our instrument. The findings are snapshots of two models at one moment and will decay as models change. The question the instrument asks—*do instructions reach the decision, or only the story about the decision?*—must be re-answered by every model generation. The durable contribution is the validated capacity to ask it.

2 MOTIVATION AND RELATED WORK

From opinion surveys to behavior. A substantial literature documents left-of-center tendencies in LLM outputs, largely by administering political-orientation questionnaires or eliciting stances on contested issues (e.g., Rozado, 2024; Feng et al., 2023; Santurkar et al., 2023; Hartmann et al., 2023; Motoki et al., 2024). Recent critiques observe that forced-choice questionnaires are themselves unstable measurements of model behavior (Röttger et al., 2024), that such measurements cannot distinguish a model’s *viewpoint preferences* from failures of *epistemic integrity*, and that directional asymmetry is not per se evidence of bias when inputs themselves are asymmetric (Rozado & Tetlock, 2026). We adopt this distinction rather than dispute it, and operationalize what it leaves unmeasured: not what models *say they believe*, but what they *do* to political framing—in the framing-theory sense of selection and salience (Entman, 1993; Chong & Druckman, 2007)—when performing realistic editorial tasks: detection, summarization, classification—on real news, with the article’s outlet and headline withheld to prevent source-prior shortcuts. Our pre-registered justified-asymmetry companion analysis (§3) directly addresses the strongest objection this literature would raise against our headline test: per-stratum intensity balance is measured (objective loaded-lexicon density; article intensity ratings; the models’ own pre-directive detection counts), with the interpretation rule fixed before data collection. Closest to our subject, Anthropic’s open-sourced political even-handedness evaluation (Anthropic, 2025) measures response symmetry to paired opinion prompts and reports the same model we study among the most even-handed available; our instruments measure framing substitution in editorial rationales on real news. These are different constructs and both results can be true of the same model—indeed their coexistence instantiates this paper’s central claim that output-symmetry and behavioral-substitution measurements can dissociate—so we position the two as complements. Of the model pairs present in both studies the orderings agree (GPT-5 is rated less even-handed there and shows the largest Lean-Right compression here, Finding 5); between-target orderings in our own tables are untested descriptives. Behavioral directional tendencies in news processing have adjacent and concurrent precedents that we position against precisely: summarizers (fine-tuned and LLM) alter article stance in over half of summaries, with roughly a quarter drifting to partisan extremes and direction dataset-dependent, under classifier-based stance measurement (Liu et al., 2024); default (no-alignment-instruction) summaries of hot-button topics by 2023-era open models drift toward Democrat-aligned framing under a classifier-distinguishability index (Vijay et al., 2025); and concurrent work reports pervasive “ideological center-collapse” in frontier-model news classification and steered generation on AllSides-labeled articles (Kennedy et al., 2026). No prior or concurrent work combines the four elements that jointly constitute this paper’s delta: substitution asymmetry measured *in rationales*, conditional on expert-rated source lean (Kennedy et al., 2026 condition on the same labels but do not measure rationale substitution); a neutrality directive manipulated against a decision-channel equivalence test; decisions separated from rationales; and reliability-gated instruments (Kennedy et al., 2026, for example, score with a single fixed LLM evaluator and note that results should be interpreted relative to it).

Faithfulness of stated reasoning. Turpin et al. (2023) showed that chain-of-thought explanations can omit the features that actually drive predictions; Lanham et al. (2023) found stated reasoning often causally irrelevant to the answer it accompanies; and under optimization pressure models learn to hide intent from stated reasoning while the behavior persists (Baker et al., 2025). Our design asks the complementary question: not whether explanations *omit* true causes, but whether an explicit instruction changes the explanation *without* changing the decision it purports to explain—and whether that rewriting has a political direction. The generation-order arms (rationale emitted before vs. after the label) probe the same post-hoc hypothesis from a second angle.

LLM judges and their validation. LLM-as-judge evaluation is known to exhibit position, verbosity, and self-preference biases (Zheng et al., 2023; Ye et al., 2025; Panickssery et al., 2024), and agreement with human raters varies widely across tasks. Reliability-with-consequence is emerging at the *instance* level—selective evaluation with provable human-agreement guarantees and escalation fallbacks (Jung et al., 2025)—but at the *instrument* level the gap is measurable: in a structured audit of a convenience sample of 25 papers from 2024–2026 that use LLM judges to score outputs and draw substantive conclusions (Appendix A), a majority (56–60%) report no inter-judge or judge–human agreement statistic of any kind, *none* (0/25; rule-of-three 95% upper bound $\sim 14\%$) states an acceptance threshold that judge quality had to meet before its scores were used, and one (4%) specifies any fallback should the judge prove unreliable. Our protocol imports the missing step from the instrument-qualification traditions of psychometrics, content analysis, and clinical-trial practice: instruments earn confirmatory

status by passing acceptance criteria fixed in advance, or they are demoted—a rule that cost this study two of its three instruments (§4.1). The gate value ($\kappa \geq 0.40$, the conventional “moderate” floor; Landis & Koch, 1977) is a *screening* criterion, not a confirmatory reliability standard: content-analysis practice requires $\alpha \geq 0.80$ for conclusions and 0.667 for tentative ones (Krippendorff, 2004), levels no LLM-judge instrument here sustains at scale—which is why claims are graded and final arbitration is human. Pre-committed quality criteria are standard in Registered Reports (Chambers & Tzavella, 2022), pre-registration is established in NLP (van Miltenburg et al., 2021), rater-level admission tests are routine in crowdsourced annotation, and judge *selection* by human agreement is practiced in benchmark construction (Dubois et al., 2024; Lambert et al., 2024)—but selection chooses the best judge post hoc, where a gate pre-commits pass/fail with a registered fallback. We are not aware of prior work in *LLM-judge evaluation* in which instrument-level gates with registered fallbacks were pre-committed in a study registration before confirmatory data collection (0/25 in the audited sample); the practice is imported—the demonstration of what it catches is new.

Construct validity in NLP evaluation. Our approach follows calls to treat evaluation instruments as measurement devices requiring validity arguments rather than as ground truth (Jacobs & Wallach, 2021; Flake & Fried, 2020). The gate-failure post-mortems (§4.1) supply concrete instances: identical rubrics resolved in opposite, internally consistent ways by two frontier judges; Cohen’s κ collapsing under extreme base rates while raw agreement stands at 87% (Gwet, 2008).

3 EXPERIMENTAL SETUP

Corpus. $N=200$ contemporary U.S. news articles on a balanced 5×8 grid: five AllSides-anchored lean classes (Left, Lean Left, Center, Lean Right, Right; 40 each) crossed with eight topical themes (25 each), 400–1500 words, per-outlet caps within lean, deterministic selection from a 10,000-article enriched pool by a released curation script. Articles were boundary- and interior-cleaned; an identity audit (id $\leftrightarrow$ url $\leftrightarrow$ text) was performed after six mismatched rows were discovered in the pilot pool (all excluded from analysis; none present in the final corpus). Models receive *article text only*—no headline, outlet, URL, or date—so that classification cannot reduce to outlet priors.

Targets and tasks. Two frontier models (Claude Sonnet 4.5; GPT-4.1) perform three tasks: **Eval A**, span-level bias detection (task lineage: Spinde et al., 2021; Hamborg et al., 2019) with per-detection explanations; **Eval B**, summarization with key facts and represented perspectives; **Eval C**, five-class lean classification with a mandatory free-text rationale. Structured JSON outputs; one rollout per cell at provider-default sampling (temperature 1.0), with a same-arm test-retest slice to anchor decoding-stochasticity floors for the equivalence results.

Conditions. Each task runs under prompt arms that differ by exactly one block (verified by programmatic diff): *baseline* (task only), *ablation* (adds bias-type schema and vocabulary; no directive), *reframing* (ablation + the neutrality directive), *full* (ablation + an attribution rule), and, for Evals A and C, a *reframing-CoT* arm that reorders generation so the rationale precedes the decision. The registered directive contrast is reframing vs. ablation.

Instruments and gates. Judge instruments are scored independently by a cross-family pair (Claude Sonnet 4.6; GPT-5) on identical inputs. Three were built: **VAR** (voice adoption: does an explanation *describe* loaded source framing or *inherit* it?), **FDC** (1–7 frame-distance coding on attribution and schema axes), and **Directional RD** (categorical: does an output’s framing substitute *leftward*, *rightward*, neither, or is there no signal—plus a strip/amplify mechanism, with quoted-evidence requirements). Each faced a pre-committed acceptance gate on held-out pilot data (excluded from all confirmatory analysis): $\kappa \geq 0.40$ with non-degeneracy conditions, one attempt, registered fallback (demotion to exploratory, per-judge reporting). Human calibration arbitrates all judged constructs—demoted and surviving alike—under a decision rule fixed before coding begins: two independent paid coders (a third adjudicating disagreements), blind to arm, target model, and judge verdicts, code a stratified $n=300$ subsample (all 58 both-judge-directional RD items; 92 stratified one-judge and neutral RD verdicts; 100 VAR verdicts including traceability-flagged and control strata; 50 articles for lean, including all 33 all-model-failure articles). H26’s construct is *sustained* if human direction labels agree with the pooled judge at $\kappa \geq 0.40$ and independently reproduce the asymmetry’s sign on the coded subsample, and *retracted to exploratory* if either condition fails; results: [pending].

Confirmatory hypotheses (three surviving tests after gate outcomes. Multiplicity follows the registered in-family treatment: BH-FDR (Benjamini & Hochberg, 1995) at $q=.05$ over all three, with H29 entering via its one-sided equivalence p (the share of bootstrap resamples below the 0.85 floor): $p_{H26}=.0014$, $p_{H28}<10^{-3}$, $p_{H29}=.005$, all significant. H29’s decision rule additionally binds on the interval itself (CI lower bound ≥ 0.85); an earlier draft presented it as interval-only, a presentation deviation logged in the registration. The headline requires all three tests—an intersection-union structure under which the correction is conservative. Full decision rules in the pre-registration): **H26**: on Eval-C rationales, $P(\text{left-substitution} \mid \text{Right-lean source}) > P(\text{right-substitution} \mid \text{Left-lean source})$; primary test judge-pooled with a per-judge sign-consistency requirement (a significant pooled result with discordant per-judge signs counts as non-confirmation). **H28**: equivalence (TOST; Lakens, 2017, $|\Delta| < 2.0$ detections/article) of Eval-A detection counts, reframing vs. ablation. **H29**: stability of Eval-C lean labels across the directive contrast (bootstrap $\kappa \geq 0.85$). H28 and H29 require no judge; H26 uses the sole gate-passing instrument.

Pre-registration and integrity. All hypotheses, thresholds, fallbacks, exclusions, and amendments are timestamped in a public git history; every instrument revision was committed *before* the data that evaluated it existed. The two demotions below were executed exactly as registered.

4 FINDINGS

4.1 FINDING 1: PLAUSIBLE JUDGE INSTRUMENTS FAIL CALIBRATION SYSTEMATICALLY—AND GATES CATCH WHAT INSPECTION DOES NOT

Finding 1 precedes Stage-2 data by design; Table 1 summarizes.

Instrument	Gate ($n=40-60$)	Outcome	Stage-2 ($n\approx 1-2k$)	Failure mode
FDC (1–7 Likert)	$q_w\text{-}\kappa = 0.109$	fail	$q_w\text{-}\kappa = 0.117$ [.106, .127]	anchor divergence: one judge 89% at ceiling; systematic offset (+2.76 at scale)
VAR (categorical, 3)	$\kappa = 0.000$	fail	$\kappa = -0.003$ [−.03, .04]	zero-variance rater (1.2% minority label at scale); raw agr. 89.5%, $AC_1=0.883$: base-rate degeneracy
Directional RD (categorical, 4)	$\kappa = 0.606$ [.28, .85]	pass	$\kappa = 0.255$ [.20, .31]	— at scale: threshold noise (neutral-vs-directional); sign agreement 93.5% (see text)

Table 1: Pre-committed reliability gates ($\kappa \geq 0.40$, one shot, registered fallbacks; cross-family judge pairs on held-out pilot items), with post-hoc large- N re-estimates from Stage-2 dual-judged data. The RD sign-agreement companion criterion passed at 100% but on a single both-directional pair.

Four observations. *First*, both failures were systematic, not noisy—and both **replicate, within the gate estimates’ intervals, at 17–50× the gate sample size** (Table 1): each judge applied an articulable, internally consistent rule that the rubric failed to adjudicate. *Second*, the pilot’s most striking apparent result—55–64% of explanations inheriting loaded source framing—was substantially a judge artifact: 44.1% of “inheriting” verdicts with extractable quoted evidence (466/1,057, of 1,225 such verdicts total) cited quoted phrases that fail verbatim source-traceability—no extracted phrase appears anywhere in the source—and the registered post-mortem records the complementary error on a further 27% (terms claimed as unquoted that were visibly quoted). The *detection* layer of this figure rests on no model’s judgment: a released script does regex quote-extraction and substring matching against the source text. The script itself illustrates the standard it serves—adversarial re-review of its first version (which reported 46.7%) found a quote-extraction defect (possessive apostrophes minting phantom quotes) and an unapplied registered exclusion (six identity-contaminated pilot articles, on which absent quotes are corpus error, not judge error); the corrected script prints the full sensitivity table. What substring matching cannot settle is *interpretation*: a traceability failure is judge error unless the phrase is a close paraphrase, a stratum the human audit of flagged verdicts (§3) quantifies. Under the corrected rubric the rate fell to 0–13% (Sonnet judge 13.3%, GPT-5 0%, on the gate slice). Ungated, this artifact was publishable. Figure 2 shows the failure texture on real items: articulate, internally consistent verdicts that pass any spot-check individually and contradict each other only when paired. *Third*, the surviving instrument differed by design

Gate protocol for LLM-judge instruments v1.1, as amended by this study’s results; template in the released artifact

1. Pair two judges from different model families on identical inputs (breaks self-preference circularity). **2. Score** a held-out pilot slice excluded from all confirmatory analysis, sized so the agreement interval’s lower bound can clear the threshold ($n \approx 40$ suffices only for structural failures). **3. Pre-commit** in a public registration, before the data exist: the acceptance criterion—interval lower bound $\geq$ threshold ($\kappa \geq 0.40$ is a screening floor, not a confirmatory standard)—plus non-degeneracy conditions (no zero-variance rater, no near-ceiling single-label marginal) and any direction-quality companions. **4. One shot:** no iterative re-anchoring against gate data. **5. Register the fallback:** failing instruments are demoted to exploratory per-judge reporting and routed to human arbitration—not deleted, not retried. **6. Report** the gate estimate with its interval wherever the instrument’s output is used, and grade claims by at-scale reliability. *Marginal cost here: two judge passes over 40–60 pilot items per instrument; it caught 2 of 3 instruments, including an artifact one decision from publication. 0/25 audited papers pre-commit any such threshold (Appendix A).*

Figure 1: The liftable protocol. Steps 2–3 carry the study’s own amendment: the original point-estimate gate was decisive for this study’s failures but only provisional for its pass (§4.1, fourth observation).

features, not luck: categorical labels rather than Likert anchors, quoted-evidence requirements for non-default verdicts, and explicit abstention semantics (κ rose from 0.233 to 0.606 under these repairs, while the same strategy could not save the Likert instrument). *Fourth*, the pass verdict itself did not survive scale unqualified: the RD gate’s point estimate failed to replicate— at $n=1,987$ the 4-label κ attenuates to 0.255, and the gate estimate’s own bootstrap interval ($[0.28, 0.85]$ at $n=40$) had always included sub-threshold values. The large- N disagreement concentrates at the neutral-vs-directional *threshold* (raw agreement 83.1%, with $\sim 88\%$ neutral verdicts mechanically depressing κ); the error type that could inject direction into H26—sign flips—occurs on 6.5% of both-flagged items (sign agreement 93.5%). Threshold noise of this kind is judge-idiosyncratic and attenuates power rather than creating asymmetry, and the H26 primary was designed for it: each judge must carry the effect independently (§3), which both did. The methodological lesson, stated precisely: **gates should bind on the interval’s lower bound or on structural-degeneracy diagnostics; a point estimate at $n \approx 40$ settles neither direction**. These two failures were decisive not because their point estimates were low but because their modes were structural (a zero-variance rater; an 89%-at-ceiling anchor) and therefore near-certain to persist; the pass was provisional precisely because it was only a point estimate. We adopt the interval-bound rule for all future instruments—and note its reflexive consequence: applied to this study it would have demoted the surviving instrument as well (gate interval $[0.28, 0.85]$), and H26 would be reported as per-judge exploratory. H26’s confirmatory status rests on the registered point rule executed as written, and the validity grading in Finding 3 absorbs exactly this tension.

4.2 FINDING 2: DECISIONS ARE EQUIVALENT ACROSS THE DIRECTIVE CONTRAST (H28, H29)

Adding the neutrality directive left both decision channels statistically equivalent to the no-directive control. **Detection counts (H28):** across $n=399$ paired article $\times$ target cells (1 of 400 lost to an unparseable output), the directive changed the number of detections by $+0.69$ per article (90% CI $[0.48, 0.90]$) against a mean of ~ 6.9 —well inside the registered ± 2.0 equivalence bound (TOST $p < .001$); equivalence in fact holds down to ± 1.0 ($p = .007$), half the registered bound. The interval excludes zero: the directive produces a small ($\sim 10\%$ relative) increase in detections—equivalent under the SESOI, not literally null. The same graded language applies to every decision-channel effect in the paper. **Lean labels (H29):** across $n=394$ paired cells (6 of 400 lost to unparseable outputs), five-class lean labels were stable across the contrast at $\kappa = 0.900$ (article-clustered bootstrap 95% CI $[0.864, 0.932]$; lower bound above the registered 0.85 floor, which binds on the interval, not the point estimate). The registered companions sharpen the picture: the 5×5 label-shift matrix is near-diagonal with per-3-class-stratum κ of 0.85–0.90 (no stratum-differential instability), while the registered span-overlap check shows the *specific spans* cited across arms overlap only partially (pooled span-level Jaccard 0.20 on exact string match; 0.23 Sonnet, 0.17 GPT-4.1)—the directive leaves the quantity and the verdicts of detection intact but not the identity of every cited span. A same-arm test-retest slice supplies the floors that make these numbers interpretable ($n=200$ article cells,

Panel A — FDC (1–7 Likert scale): same rationale, opposite ends of the scale. gate $qw-\kappa = 0.109$; replicated 0.117 at $n=1,987$

Rationale under judgment (abridged): “The article demonstrates a predominantly left-leaning framing [...] using emotionally charged language that aligns with Democratic messaging priorities. The article opens by highlighting ‘skyrocketing health care costs’ [...] The article’s structure suggests Republican obstruction [...] while portraying Democrats as advocating for vulnerable Americans.”

Claude Sonnet 4.6 attribution 2, schema 2

“The reasoning *adopts the source’s framing as its own analytical lens* [...] its own summary re-uses charged terms like ‘obstruction’ and ‘vulnerable Americans’ without attribution.”

GPT-5

attribution 5, schema 7

“It *quotes and marks the article’s loaded language* [...] but also adds some unquoted evaluative phrasing (e.g., ‘Republican obstruction’), while consistently analyzing the framing [...] rather than reasoning within it.”

Panel B — VAR (categorical): opposite verdicts, and a judge that argues with itself. gate $\kappa = 0.000$; replicated -0.003 at $n=996$; $AC_1 = 0.883$

Explanation under judgment: “The author presents his beliefs about private prisons in a definitive way, as though they are established facts, without indicating they are opinions and without qualifying language.”

Claude Sonnet 4.6 verdict: **inheriting**

“[the explanation] fails to attribute or quote the source’s framing with distancing markers—however, since there is no loaded source phrasing inherited here [...] *making this describing*.” (The stated rationale concludes *describing*; the verdict field records *inheriting*.)

GPT-5

verdict: **describing**

“It attributes the stance to the author and analyzes the framing without reusing the source’s loaded phrasing as fact.”

Figure 2: **Why gating was necessary.** Two real Stage-2 items from the instruments that failed their pre-committed reliability gates. In Panel A the judges *agree on the facts*—both note the unquoted “Republican obstruction”—yet score opposite ends of the 1–7 scale: anchor divergence, not factual disagreement. In Panel B the category boundary is ill-defined enough that one judge’s verdict contradicts its own stated rationale. Every verdict shown is fluent and individually plausible; the unreliability is visible only when verdicts are paired across judge families and scored against a criterion fixed in advance. Spot-checking cannot catch this; gates did.

identical prompts re-run; executed for the Sonnet target only, a partial execution of the registered both-target re-run): count difference $+0.02$ (SD 2.2), span Jaccard 0.269, label $\kappa = 0.960$. In the matched-target comparison, cross-arm span overlap (0.228) sits near the stochasticity ceiling (0.269)—most span churn is decoding noise, not directive effect—while label stability across the directive (0.915) against the 0.960 retest ceiling indicates a small real directive effect on labels, consistent with the graded reading throughout.

Equivalent counts could still hide asymmetric *content*—flagging right-coded language as bias more readily than left-coded language—so we checked: right-coded lexicon terms present in a source are flagged at 36.0% vs. 38.5% for left-coded terms (difference $-2.5pp$, clustered bootstrap CI $[-14.0, +9.3]$), stable across both targets and all five arms. This matters because it finds no evidence for the last decision-side loophole—though the interval is wide, so it excludes large content asymmetries ($>\sim 10pp$) rather than all of them: the models’ detection behavior shows no detectable political skew in either volume or content, which is what makes the rationale-side drift (Finding 3) a dissociation rather than one symptom of a general bias. The only content structure found is *lean-congruence*—models preferentially flag the article’s own side’s language ($+11.1pp$, marginal)—which is what a competent bias detector should do.

4.3 FINDING 3: RATIONALES DRIFT, DIRECTIONALLY (H26)

On the same rollouts whose decisions were equivalent, the models’ free-text rationales substituted political framing asymmetrically. Pooling both judges with article-level clustering (80 Right-lean vs. 80 Left-lean articles): left-coded substitution on right-leaning sources occurred on 11.0% of

classifiable rationales, against 5.8% for the mirror-image substitution on left-leaning sources—a +5.2pp asymmetry ($1.9\times$; one-sided $t = 3.05$, $p = .0014$, significant under BH-FDR in the three-test family). The pre-registered robustness requirement was exceeded: both judge families were *independently* significant in the predicted direction (Sonnet +4.0pp, $p = .006$; GPT-5 +6.4pp, $p = .004$)—sign agreement was required, but each judge carries the result alone. In raw counts (of 1,579 classifiable judge-verdicts on right-leaning and 1,586 on left-leaning sources), right-leaning sources drew 173 left-coded against 48 right-coded substitutions, left-leaning sources 92 right-coded against 23 left-coded: both source classes are re-framed *toward the center*, right-leaning sources more strongly. Instrument coverage was high (no-signal verdicts 1–3% per arm; 0.3–0.4% on partisan strata, 8.4% on Center—the procedural-copy case the label was designed for), and the observed magnitude matched the post-anchoring power projection ($1.9\times$ vs. the pilot’s judge-lenient $2\text{--}3\times$). Item-level inter-judge reliability at scale is fair rather than moderate ($\kappa = 0.255$; §4.1), driven by threshold rather than sign disagreement (sign agreement 93.5%); because such noise is judge-idiosyncratic, it works against detection, and the per-judge independent significance bounds its influence on the conclusion.

Three characterizations sharpen the result. **It is not misperception:** the drift persists when perception is demonstrably correct—restricting *both* strata to articles whose lean the model classifies exactly at baseline, left-substitution on the 39 correctly-read right-lean articles is 19.7% against 7.5% for the mirror on the 48 correctly-read left-lean articles ($p = .0012$; unrestricted mirror 7.3%)—so the default is a property of expression, not of perception. (The correct-vs-misclassified contrast itself, 19.7% vs. 15.1%, is underpowered: the minimum detectable difference at 80% power is 13.4pp, so no dependence claim is made in either direction.) **It is model-specific:** decomposing by target (exploratory; the registered reporting was per-judge), the asymmetry is strong in Claude Sonnet 4.5 (17.4% vs. 7.3%, +10.1pp, one-sided $p = 10^{-4}$) and not detected in GPT-4.1 (4.4% vs. 4.3%, +0.1pp, 95% CI $[-3.0, +3.2]$), whose rationales carry little directional substitution of either sign; the formal between-target interaction is +10.0pp (article-paired bootstrap: the shared 80+80 articles are resampled once per draw, each carrying both targets’ rates jointly; 95% CI $[+4.7, +15.4]$, $p \approx .0002$), so the difference itself is significant, not merely the contrast of a significant with a non-significant result. On the ratio scale (robust to the targets’ different directional-content base rates), Sonnet’s center-ward substitution ratio is 2.4 vs. GPT-4.1’s 1.0. The registered pooled test is thus carried by one of the two targets; the confirmed claim is correspondingly a directional default in *a* frontier model, not in frontier models per se. The model-specificity itself is informative: the behavior is not forced by the task or the corpus. **Its mechanism is additive:** among directional verdicts on right-leaning sources, left-substitution operates by *amplification*—introducing left-coded framing absent from the source—three times as often as by stripping right-coded terms (119 vs. 39), whereas rightward moves on left-leaning sources are predominantly strips (53 vs. 36). The default, where present, operates chiefly by introducing left-coded terms absent from the source rather than by deleting right-coded ones. (The mechanism subfield was not separately gate-validated; raw cross-family agreement on it among both-directional items is 79%, 49/62—the characterization is descriptive.)

Four robustness checks, each targeting a distinct way the result could be an artifact. *Distributional assumptions:* a 10,000-draw article-relabeling permutation test (post-registration robustness) gives $p \leq 10^{-4}$ (0 of 10,000 permuted statistics reached the observed value at the registered seed; add-one corrected $p = 10^{-4}$), so the parametric primary is, if anything, conservative. *Stratum choice* (Sonnet target, where the effect resides): the asymmetry is significant at both extremity levels separately (Right vs. Left +11.8pp, $p = .0008$; Lean Right vs. Lean Left +8.4pp, $p = .019$; pooled across targets the extreme pair holds at +7.3pp, $p = .0015$, while the lean pair does not, +3.1pp, $p = .11$)—a level shift across the right half of the spectrum, not an artifact of extreme outlets. *Topic* (Sonnet target): the asymmetry is positive in all 8 of 8 themes (range +4 to +20pp; pooled across targets, 7 of 8, with foreign_defense at −0.9pp), so it is a property of the model’s treatment of right-coded framing generally, not of one charged topic. *Judge family:* a judge $\times$ target difference-in-differences finds no same-family favoritism (n.s., and scale-dependent in sign); decisively, the *cross-family* judge finds the larger effect, so any residual own-family leniency could only have shrunk the result, not produced it.

Measurement-validity status of this result. The instrument passed its registered gate, but three facts qualify it and two independent lines of evidence sustain it. Qualifying: item-level κ at scale is 0.255, below the 0.40 gate value the instrument passed at $n=40$; the gate’s sign-agreement criterion, though passed at 100%, was satisfied on a single both-directional item pair; and inter-judge

threshold disagreement is itself stratum-differential (one-judge-flags rates 17.8% on right-lean vs. 10.8% on left-lean items, $p < .001$)—the pattern a manufactured-asymmetry account would require. Sustaining: first, the asymmetry survives restriction to the maximum-reliability subset, the 58 items where *both* judges agree on direction (+2.8pp, one-sided $p = .005$), which discards every disputed verdict; second, the originally registered *judge-free* instrument—the deterministic paired lexicon, a descendant of paired partisan-phrase slant measurement (Gentzkow & Shapiro, 2010), applied to the same rationale text—independently finds the same-sign asymmetry (Sonnet target: added left-coded terms on right-lean sources 21.8% vs. the mirror 14.3%, +7.5pp, $p = .04$; pooled targets +5.8pp, $p = .03$), and rationale lengths are balanced across strata (Sonnet: 172.7 vs. 178.1 words), so neither judge priors nor text volume explains the direction. The lexical corroboration is soft: it decomposes into two individually non-significant components (slightly higher coded-vocabulary engagement on right-lean articles, and slightly more left-adding conditional on engagement), so it confirms the sign, not the magnitude, of the judge-measured effect; and because coded-vocabulary engagement is itself part of the generated rationale—potentially raised by the very default being measured—the decomposition is post-outcome stratification, descriptive rather than causal. The result is additionally robust to outlet-level clustering (CI [+0.012, +0.092]). Magnitudes should be read as a bracket rather than a point: the both-judge-agreed floor is +2.8pp against the pooled +5.2pp, and absolute rates are judge-dependent (Sonnet judge 7.4% vs 3.4%; GPT-5 judge 14.5% vs 8.2%). Distribution shift between the pilot gate slice and the deployment population is a live explanation for the κ attenuation, alongside small- n noise. We therefore report H26 as confirmed under its registered procedure and corroborated by a judge-free instrument, with final construct arbitration assigned to human coding.

4.4 FINDING 4: THE ASYMMETRY IS NOT EXPLAINED BY INPUT INTENSITY (D-JA COMPANION)

All three pre-registered intensity indicators fail to account for the asymmetry. Computed pre-collection: objective loaded-lexicon density is balanced across Left and Right strata (1.73 vs. 1.93 per 1k words; $d=0.10$, $p=.55$); the article-intensity ratings (LLM-generated, with a registered circularity caveat: a rater sharing the measured default would under-rate right-leaning intensity, producing exactly this pattern) are imbalanced *against* the predicted direction (Left articles rated more intense). Under a strip-proportional account this imbalance is conservative; under the observed amplify-dominant mechanism it is more nearly neutral, which is why the two model-independent indicators (lexicon density, and the detection counts below) carry the primary weight. Computed from Stage-2 data: the models’ own pre-directive assessment of bias density is balanced—baseline detections per article 5.96 (Left) vs. 6.38 (Right), $d=0.13$, $p=.40$. (Detection volume is meanwhile well-calibrated to bias presence: partisan articles draw 2.3× the detections of Center articles, a discriminant-validity check on the detection task itself.) Under the pre-registered interpretation rule, H26 is therefore reported as a *directional default in substitution behavior*, not as a proportionate response to asymmetric inputs.

4.5 FINDING 5 (EXPLORATORY): CLASSIFICATION CAPABILITY, AND A THIRD DIRECTIONAL SIGNATURE

Five-class lean classification against AllSides labels is moderately accurate for all four frontier models in the study (targets: 53–66% exact, 86–94% within one class, vs. 20% chance; judges: 61–65% exact), providing the capability context for H29’s stability result. The *errors*, however, are not directionally neutral. Signed error by true stratum (negative = classified left of the expert label):

Model	Left	Lean Left	Center	Lean Right	Right
Claude Sonnet 4.5 (target)	+0.05	−0.38	−0.55	− 0.75	−0.30
GPT-4.1 (target)	+0.18	+0.15	−0.22	− 0.64	−0.42
Claude Sonnet 4.6 (judge)	+0.08	−0.18	−0.35	− 0.65	−0.38
GPT-5 (judge)	+0.18	+0.12	−0.38	− 1.05	−0.65

Table 2: Signed classification error (in 5-class steps) by true stratum; targets at baseline, judges from the LCA instrument. Symmetric center-regression would produce mirror-image errors; instead the right half of the spectrum is compressed toward Center while the left half is classified nearly accurately.

All four models—two families, targets and judges alike—absorb “Lean Right” toward Center by 0.6–1.1 classes while classifying left-of-center strata nearly accurately. This was not a pre-registered test and is reported as a descriptive observation, but its structure is notable: it is a third, independent measurement channel (classification error, alongside framing substitution and rationale voice) exhibiting the same directional compression of right-coded content. Two caveats: the analysis inherits AllSides’ outlet-anchored labels as ground truth (the standard caveat for outlet-derived corpora: Baly et al., 2020), so the pattern could partly reflect article-level deviation from outlet ratings—though uniform label noise cannot produce the observed left/right asymmetry—and the H26 result is independent of this channel by construction (the RD judge never sees or produces lean labels). Concurrent work independently observes the collapse itself: Kennedy et al. (2026) report pervasive ideological center-collapse in nine frontier models (including one target and one judge studied here) classifying and generating against the same AllSides label ontology. Their analysis does not decompose the classification-stage collapse by side, so Table 2’s left/right decomposition is a refinement rather than a replication—though their generation-stage observation that models hold the right pole less accurately than the left under steering is directionally consonant with the compression here. Their scoring relies on a single fixed LLM evaluator (a dependence their limitations flag), the instrument class Finding 1 gates. The models’ self-reported confidence adds a mitigating nuance: the Lean Right→Center misclassification is their *lowest*-confidence error cell (.82 vs. .85 on other errors, $p < 10^{-6}$; Sonnet .77). The compression is a hesitant boundary failure, not confident assimilation—which matters practically, because it means the confidence signal already flags the failure and could be used to route exactly these cases to review. Classification error is, moreover, substantially article-driven: 33 of 198 articles (2 of 200 lack labels from at least one model) are misclassified by *all four* models ($7.8\times$ the rate independent errors would produce; mean pairwise correctness correlation 0.49), and correctly-classified articles carry nearly twice the rated intensity of misclassified ones (3.17 vs. 1.82)—a shared-difficulty structure consistent with article-level ambiguity or outlet-to-article label error. Article-level human coding adjudicates, prioritizing the all-model-failure set.

4.6 FINDING 6 (DESCRIPTIVE): BOUNDARIES OF THE EFFECT

Three registered descriptive analyses bound the phenomenon. **Text type:** on extractive summaries (Eval B), the *objective* lexicon instrument—deterministic paired left/right term lists, no LLM judge—finds no retention asymmetry (own-coded terms retained at 0.42 on Right-lean vs. 0.40 on Left-lean sources, $p = .63$). Either the directional default is specific to free-voice analytical text (rationales drift; summaries stay anchored to source content), or the lexicon’s coverage ($\sim 7\%$ of framing-bearing language) lacks power; we report both readings. One summary-channel candidate does appear in a different field: summaries credit right-lean articles with ~ 0.5 fewer represented perspectives (3.42 vs. 3.94, $p = .006$, same direction in both targets, attenuated but persistent under a word-count control) — unverifiable as model behavior without ground-truthing how many sources each article actually quotes, and therefore flagged for the human coding rather than claimed. **Generation order** (D-HCoT): emitting the rationale *before* the decision shifts detection counts by -0.66 per article ($CI_{90} [-0.89, -0.43]$)—nearly the mirror image of the directive’s $+0.69$ —and perturbs $\sim 22\%$ of lean labels (vs. $\kappa = 0.90$ stability across the directive contrast at fixed order), with rationales lengthening from 145 to 214 words. Instructing the rationale channel does not move decisions, but *re-ordering* it does, modestly: the rationale is not purely epiphenomenal, and “post-hoc” should be read as a claim about the directive’s locus of effect, not about architecture. **Model:** as reported above, the directional component is confined to one of the two targets. **Anatomy** (exploratory, Sonnet target): the default is article-anchored rather than diffuse—the twenty most-affected articles carry 70% of all left-substitution verdicts, sixteen articles drift in three or more of the five arms (persistence exceeding an independence baseline), and the persistent triggers skew toward higher-intensity, full-Right sources from combative outlets. Its lexical content is largely *institutional renaming of contested referents*: the most-added left-coded terms are “pro-choice,” “criminal justice reform,” “reproductive rights,” “undocumented,” and “gender-affirming care”—style-guide normalization whose vocabulary is left-coded on precisely these topics (the lexicon codes term presence, not stance, so this table characterizes rather than adjudicates). This vocabulary profile admits a rival reading of the whole phenomenon: asymmetric conformity to a mainstream journalistic house register that is politically coded on precisely these contested topics—an account that would produce the additive mechanism, the both-classes-centerward flow, and the model-specificity (vendors’ style tuning differs) without any political processing. The registered discriminating test (mirrored-vocabulary turnabout counterfactuals) was deferred as future work in the registration and has not been run; until it runs, “directional default”

names measured behavior relative to expert-rated source lean, not an adjudicated cause. And it operates without self-signal: within the same article, the model’s confidence is identical on drifted and non-drifted rationales ($\Delta = +0.002$, $p = .77$; an unpaired gap of $+0.027$ reflects composition—drift concentrates on the articles the model handles most confidently), so no introspective probe of the model would flag the substitution as it happens. **Instruction:** the drift needs no prompt to exist and no prompt removes it. It is already present at baseline ($+9.4\text{pp}$, $p = .007$, Sonnet target) and remains significant under every arm—bias taxonomy, attribution rule, neutrality directive, reasoning-first ordering ($+7.5$ to $+13.1\text{pp}$). The nominally largest value, under the neutrality directive, is attributable to that arm’s longer rationales (more text, more codable framing): controlling for length, the directive’s apparent elevation disappears and reverses. The meaning is double-edged: no intervention we tested *causes* the drift, and none *cures* it—it is a resting-state property of the model, unmoved by every prompt intervention we tested. Together these boundaries argue for the paper’s central methodological point: each of these distinctions is invisible to any evaluation that reads model self-reports at face value, and none was expensive to measure—given validated instruments.

5 DISCUSSION AND LIMITATIONS

What the confirmed dissociation means. The confirmed structure is two-sided: decisions were statistically equivalent across the directive contrast (H28, H29), while rationales carry a directional default that exists with no instruction and that no tested instruction removes (H26; Finding 6). The two channels answer to different levers—that is the dissociation—but the confirmed rationale-side result is *directive-independent*. Whether the directive additionally rewrites rationales to *perform* neutrality was the question assigned to the instruments that failed their gates; the surviving evidence on it is exploratory (rationales lengthen by $+18$ to $+65$ words under the directive and by ~ 160 under reasoning-first ordering; the demoted voice instrument’s single-judge trend points the same way but, under the registered standard, carries no evidential weight). For practitioners, the confirmed components alone carry the consequence: an audit that read these models’ explanations would have had no behavioral change to detect (the directive produced none), and no self-report trace of the default to find (drifted and non-drifted rationales carry identical confidence within the same article; Finding 6). Behavioral audits—counting decisions under controlled contrasts—are not a refinement of explanation-reading but a different measurement, and here the only one of the two that could have found either result. We demonstrate one confirmed instance, not a universal law; but the existence proof is enough to make behavioral audits a necessary complement wherever self-reports are currently read as evidence—e.g., general-purpose-model evaluations under the EU AI Act (European Parliament and Council of the European Union, 2024), or the 2025 U.S. procurement criterion of LLM “ideological neutrality” (Executive Office of the President, 2025; Office of Management and Budget, 2025)—whose implementation guidance accepts system-prompt disclosure as compliance evidence—the latter exactly the setting where an ungated neutrality audit could mint policy-consequential artifacts. Our audit framing follows the internal-audit and named-system traditions (Raji et al., 2020; Raji & Buolamwini, 2019).

Design rules for news-facing AI products. The findings were collected as an audit, but each maps onto a concrete design rule for the product class named in §1—bias checkers, balanced-news summarizers, and media-literacy tools anchored to expert outlet ratings. (1) *Ship decisions; gate explanations.* The decision channels measured deployment-grade: detection volume tracks bias presence (partisan articles draw $2.3\times$ the detections of Center articles) with no detectable political skew in flagged content, and lean labels are stable—so span-level detections and quoted evidence are the safe surface, while free-voice explanatory prose is where the directional default lives and belongs behind a filter. (2) *A neutrality directive is not a mitigation.* It left decisions equivalent and the default untouched; a product claiming neutrality because its system prompt requests it commits precisely the self-report error this paper measures. (3) *Prefer extractive summarization.* Summaries showed no retention asymmetry (Finding 6)—the safer generative surface—but perspective enumeration needs auditing: right-lean articles were credited with ~ 0.5 fewer represented perspectives. (4) *Route the known failure cell to humans.* All four models compress Lean-Right toward Center and it is their lowest-confidence error cell (Finding 5), so a confidence threshold routes exactly these cases into the human editorial review that expert-rating platforms already operate; and because the 33 all-model-failure articles show the residual ambiguity is article-level, human arbitration—not a larger model—is the fix. (5) *Post-filter with the lexicon.* The judge-free paired lexicon detects the substitution deterministically at zero marginal cost; as an output filter it flags institutional-renaming

rewrites (the highest-frequency drift vocabulary) before display, and the drift’s concentration—twenty articles carry 70% of it—makes risk targeting practical. (6) *External QA only; re-audit per upgrade.* Drifted rationales carry no confidence self-signal (Finding 6), so a product cannot self-monitor through model confidence; any LLM-judge QA in its pipeline inherits Finding 1’s gate requirement; and the \$150 marginal re-run prices the audit as a per-model-upgrade regression test rather than a one-off study. These rules are the practical content of the dissociation: the audit tells a builder which channels of a bias-analysis product current models can already power, and which still need gates, filters, or humans.

What the gate failures mean regardless of Stage-2 outcomes. The measurement result stands on its own: two frontier judges given the same plausible rubric produced near-zero chance-corrected agreement twice, the failures were invisible to output inspection, and one of them had already minted an attractive, wrong number. We suggest the field’s burden of proof should invert: an LLM-judge number without a reliability argument should be treated as unvalidated instrument output, not as a finding. Unvalidated does not mean false—the demand is for reliability arguments, not a license to dismiss judge-measured findings; this study’s own directional result survives both a gated judge instrument and a judge-free one.

Limitations. (1) *Scope:* two closed-weight models, U.S. politics, English news; the pattern’s generality is untested (the pipeline re-runs on additional targets for marginal cost). (2) *No mechanism:* we establish behavioral dissociation, not its cause; generation-order arms probe but cannot settle it. (3) *Judge mediation:* H26 rides on one validated instrument; cross-family sign agreement and human calibration ([results]) bound, but do not eliminate, shared-judge-bias risk. (4) *Gate statistics:* κ is base-rate sensitive; the VAR post-mortem illustrates the κ paradox (high raw agreement, degenerate marginals), which is why gates included non-degeneracy conditions and why demotions route to human arbitration rather than deletion. (5) *Construct floor:* under the corrected rubric, explanation-level framing inheritance appears rare; whether that reflects model behavior or rubric strictness awaits human coding. (6) *One rollout per cell;* sampling variance is handled across articles, not within. (7) *AI-assisted research loop:* the audits that detected judge errors were themselves AI-executed. Their findings are secured by deterministic re-verification where possible (released scripts), but no human has yet independently coded any judged item; the pending human calibration is therefore load-bearing for the judged claims, and the paper should be read with that dependency explicit.

6 CONCLUSION

Measuring whether neutrality directives change what frontier models do with politically framed news turned out to require, first, instruments that could be trusted to measure anything at all. The resulting study contributes a confirmed, pre-registered dissociation between decisions and rationales (decisions equivalent under the directive; rationales carrying a directional default the directive neither causes nor removes), a directional characterization of framing substitution on real news that survives a justified-asymmetry discipline (1.9× toward-center asymmetry against right-leaning sources, independently significant under both judge families, localized to one of the two target models and operating by additive re-coding)—confirmed under its registered procedure, with construct arbitration pending human coding—and, independent of either, a demonstrated protocol under which evaluation instruments earn trust before they are believed. The findings are dated the moment new models ship. The capability to re-derive them—on a regenerated corpus, with re-gated instruments, for marginal API costs of roughly a conference registration (the validation layers, larger gate slices and human arbitration, are priced separately)—is the contribution we expect to last. For builders of news-facing AI, the same package is a qualification test: it answers, per model generation, which channels of a bias-analysis product are deployable and which still require gates, filters, or human review.

ETHICS STATEMENT

This study evaluates model behavior on politically contested material; we have tried to hold it to the epistemic standard we advocate. The research question is directional symmetry, not the vindication of any political position; the corpus is balanced by construction against expert ratings; the loaded-language lexicon pairs left- and right-coded terms; judge prompts were audited for direction-priming and example asymmetries, with two such asymmetries found and corrected pre-data. We adopt the

distinction between viewpoint preference and epistemic failure and explicitly do not equate measured asymmetry with “bias”: our pre-registered companion analysis tests whether asymmetric treatment tracks asymmetric inputs, and our claims are conditioned on its outcome. Misuse risks run in two directions—overreading a directional default as partisan intent, and weaponizing “neutrality” audits to punish accurate asymmetry (Rozado & Tetlock, 2026); our reporting rules (per-judge estimates, intensity normalization, equivalence tests reported alongside directional ones) are designed against both. News text is used under research norms with per-outlet caps; article text is redistributed only as required for reproduction, with source URLs and the deterministic curation script provided in lieu of bulk redistribution where licensing requires. Planned human annotation involves paid coders rating published news text (no personal data, no vulnerable populations); compensation and consent procedures follow institutional guidelines. This study was conducted with substantial AI assistance (an Anthropic-family assistant contributed to instrument drafting, analysis code, and adversarial audits), a meta-level circularity we address structurally: every load-bearing correction—the artifact rate, count reconciliations, reliability statistics—rests on deterministic, committed, re-runnable code rather than any model’s judgment; the claims that do rest on model judgment (the directional labels) are exactly those assigned to human arbitration; and the judging itself was cross-family by design. The division is disclosed so readers can weight each claim by its verification path. Model-specific results will be shared with both developers prior to public release, following named-system audit practice (Buolamwini & Gebru, 2018; Raji & Buolamwini, 2019). All API costs (~\$200 study total against the registration’s conservative ~\$775 budget—batch discounts, prompt caching, and demotion-reduced judging account for the difference; ~\$150 marginal re-run, excluding validation layers) and the full audit trail of instrument failures are disclosed to make the study’s error-correction process, not only its conclusions, inspectable.

REPRODUCIBILITY STATEMENT

The pre-registration (hypotheses, thresholds, gates, fallbacks, and every amendment) is timestamped in a public git history, with instrument revisions committed before the data that evaluated them existed. We release: the deterministic corpus curation script and corpus manifest, all judge prompts (versioned, with the full revision history), the runner and analysis code, gate validation data, batch manifests, per-call cost accounting, and a power/precision table for all three confirmatory tests (the registered deliverable, computed post hoc and labeled as such: TOST power $> .99$ at the registered bound, $.80$ at the ± 1.0 sensitivity bound; $.81$ probability of demonstrating the H29 floor at the observed κ). A regeneration of the full study on new articles or new target models requires one configuration change and ~\$150 in batched API calls. A minimum-valid-use rule accompanies the release: results produced without freshly passed gates (and, for judged constructs, without the arbitration layer) are not findings of this instrument, and the analysis code prints an UNGATED---NOT A FINDING banner when gate artifacts are absent. [Repository and OSF links after de-anonymization.]

A JUDGE-VALIDATION PREVALENCE SURVEY

Twenty-five papers (2024–2026) that use LLM judges to score model outputs and draw substantive conclusions were located via arXiv/ACL Anthology search and coded on five binary fields from their published text; two additional candidates were excluded as out-of-scope (one scores with trained reward models rather than LLM judges; one is a judge-methods paper). Coding was a quote-anchored single pass, AI-assisted, with the full per-paper table (URLs, quoted bases) released; human dual-coding of all 25 is [pending].

REFERENCES

- Anthropic. Measuring political bias in Claude. Anthropic research report, November 2025. URL <https://www.anthropic.com/news/political-even-handedness>. Paired-prompts even-handedness evaluation; open-sourced at <https://github.com/anthropics/political-neutrality-eval>.
- Bowen Baker, Joost Huizinga, Leo Gao, Zehao Dou, Melody Y. Guan, Aleksander Madry, Wojciech Zaremba, Jakub Pachocki, and David Farhi. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation, 2025. URL <https://arxiv.org/abs/2503.11926>.

Practice	Yes	No/Unclear	Rate
Reports any agreement statistic	10	15	40%
Uses judges from ≥ 2 model families	9	16	36%
Validates judge against human labels	14	11	56% ^a
Pre-commits an acceptance threshold	0	25	0%
Pre-specifies a fallback if unreliable	1	24	4%

Table 3: Prevalence of judge-validation practices ($N=25$). ^a3–4 of the 14 are qualitative-only; quantitative validation is $10/25 = 40\%$. Convenience sample over-representing benchmark-construction papers, which validate most — the no-statistic rate is therefore likely an underestimate for the broader literature. Sharpest sub-pattern (post-hoc observation, not a pre-specified field): all 11 papers that *consume* judged evaluations (rather than construct them) report zero agreement statistics.

Ramy Baly, Giovanni Da San Martino, James Glass, and Preslav Nakov. We can detect your bias: Predicting the political ideology of news articles. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 4982–4991, Online, 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.404. URL <https://aclanthology.org/2020.emnlp-main.404/>.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x. URL <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.

Joy Buolamwini and Timnit Gebru. Gender shades: Intersectional accuracy disparities in commercial gender classification. In Sorelle A. Friedler and Christo Wilson (eds.), *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, volume 81 of *Proceedings of Machine Learning Research*, pp. 77–91. PMLR, 2018. URL <https://proceedings.mlr.press/v81/buolamwini18a.html>.

Christopher D. Chambers and Loukia Tzavella. The past, present and future of Registered Reports. *Nature Human Behaviour*, 6(1):29–42, 2022. doi: 10.1038/s41562-021-01193-7. URL <https://doi.org/10.1038/s41562-021-01193-7>.

Dennis Chong and James N. Druckman. Framing theory. *Annual Review of Political Science*, 10: 103–126, 2007. doi: 10.1146/annurev.polisci.10.072805.103054. URL <https://doi.org/10.1146/annurev.polisci.10.072805.103054>.

Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. Length-controlled AlpacaEval: A simple way to debias automatic evaluators, 2024. URL <https://arxiv.org/abs/2404.04475>. Accepted at COLM 2024.

Robert M. Entman. Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, 43(4):51–58, 1993. doi: 10.1111/j.1460-2466.1993.tb01304.x. URL <https://doi.org/10.1111/j.1460-2466.1993.tb01304.x>.

European Parliament and Council of the European Union. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, OJ L, 2024/1689, 12.7.2024, 2024. URL <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>. CELEX: 32024R1689.

Executive Office of the President. Preventing woke AI in the federal government. Executive Order 14319, 90 FR 35389–35391, July 2025. URL <https://www.federalregister.gov/documents/2025/07/28/2025-14217/preventing-woke-ai-in-the-federal-government>. Signed July 23, 2025; published in the Federal Register July 28, 2025 (document no. 2025-14217).

Shangbin Feng, Chan Young Park, Yuhan Liu, and Yulia Tsvetkov. From pretraining data to language models to downstream tasks: Tracking the trails of political biases leading to unfair NLP models. In

- Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 11737–11762, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.656. URL <https://aclanthology.org/2023.acl-long.656/>.
- Jillian Fisher, Ruth E. Appel, Chan Young Park, Yujin Potter, Liwei Jiang, Taylor Sorensen, Shangbin Feng, Yulia Tsvetkov, Margaret E. Roberts, Jennifer Pan, Dawn Song, and Yejin Choi. Political neutrality in AI is impossible—but here is how to approximate it, 2025. URL <https://arxiv.org/abs/2503.05728>.
- Jessica Kay Flake and Eiko I. Fried. Measurement schmeasurement: Questionable measurement practices and how to avoid them. *Advances in Methods and Practices in Psychological Science*, 3(4):456–465, 2020. doi: 10.1177/2515245920952393. URL <https://doi.org/10.1177/2515245920952393>.
- Deep Ganguli, Amanda Askell, Nicholas Schiefer, Thomas I. Liao, Kamilė Lukošiuūtė, Anna Chen, et al. The capacity for moral self-correction in large language models, 2023. URL <https://arxiv.org/abs/2302.07459>.
- Matthew Gentzkow and Jesse M. Shapiro. What drives media slant? Evidence from U.S. daily newspapers. *Econometrica*, 78(1):35–71, 2010. doi: 10.3982/ECTA7195. URL <https://doi.org/10.3982/ECTA7195>.
- Kilem Li Gwet. Computing inter-rater reliability and its variance in the presence of high agreement. *British Journal of Mathematical and Statistical Psychology*, 61(1):29–48, 2008. doi: 10.1348/000711006X126600. URL <https://bpspsychub.onlinelibrary.wiley.com/doi/10.1348/000711006X126600>.
- Felix Hamborg, Karsten Donnay, and Bela Gipp. Automated identification of media bias in news articles: an interdisciplinary literature review. *International Journal on Digital Libraries*, 20(4):391–415, 2019. doi: 10.1007/s00799-018-0261-y. URL <https://doi.org/10.1007/s00799-018-0261-y>.
- Jochen Hartmann, Jasper Schwenzow, and Maximilian Witte. The political ideology of conversational AI: Converging evidence on ChatGPT’s pro-environmental, left-libertarian orientation, 2023. URL <https://arxiv.org/abs/2301.01768>.
- Abigail Z. Jacobs and Hanna Wallach. Measurement and fairness. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT ’21)*, pp. 375–385. ACM, 2021. doi: 10.1145/3442188.3445901. URL <https://doi.org/10.1145/3442188.3445901>.
- Jaehun Jung, Faeze Brahman, and Yejin Choi. Trust or escalate: LLM judges with provable guarantees for human agreement. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL <https://openreview.net/forum?id=UHPnqSTBPO>.
- Molly Kennedy, Ali Parker, Yihong Liu, and Hinrich Schütze. Left, right, or center? Evaluating LLM framing in news classification and generation, 2026. URL <https://arxiv.org/abs/2601.05835>. arXiv preprint; concurrent work.
- Tomek Korbak, Mikita Balesni, Elizabeth Barnes, Yoshua Bengio, Joe Benton, Joseph Bloom, et al. Chain of thought monitorability: A new and fragile opportunity for AI safety, 2025. URL <https://arxiv.org/abs/2507.11473>.
- Klaus Krippendorff. Reliability in content analysis: Some common misconceptions and recommendations. *Human Communication Research*, 30(3):411–433, 2004. doi: 10.1111/j.1468-2958.2004.tb00738.x. URL <https://doi.org/10.1111/j.1468-2958.2004.tb00738.x>.
- Daniël Lakens. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362, 2017. doi: 10.1177/1948550617697177. URL <https://doi.org/10.1177/1948550617697177>.

- Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James V. Miranda, Alisa Liu, Nouha Dziri, Shane Lyu, Yuling Gu, Saumya Malik, Victoria Graf, Jena D. Hwang, Jiangjiang Yang, Ronan Le Bras, Oyvind Taffjord, Chris Wilhelm, Luca Soldaini, Noah A. Smith, Yizhong Wang, Pradeep Dasigi, and Hannaneh Hajishirzi. Tulu 3: Pushing frontiers in open language model post-training, 2024. URL <https://arxiv.org/abs/2411.15124>.
- J. Richard Landis and Gary G. Koch. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174, 1977. doi: 10.2307/2529310. URL <https://www.jstor.org/stable/2529310>.
- Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, et al. Measuring faithfulness in chain-of-thought reasoning, 2023. URL <https://arxiv.org/abs/2307.13702>.
- Fengyuan Liu, Nouar AlDahoul, Gregory Eady, Yasir Zaki, and Talal Rahwan. Self-reflection makes large language models safer, less biased, and ideologically neutral, 2025. URL <https://arxiv.org/abs/2406.10400>.
- Yuhan Liu, Shangbin Feng, Xiaochuang Han, Vidhisha Balachandran, Chan Young Park, Sachin Kumar, and Yulia Tsvetkov. P³Sum: Preserving author’s perspective in news summarization with diffusion language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 2154–2173, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.119. URL <https://aclanthology.org/2024.naacl-long.119/>.
- Fabio Motoki, Valdemar Pinho Neto, and Victor Rodrigues. More human than human: measuring ChatGPT political bias. *Public Choice*, 198:3–23, 2024. doi: 10.1007/s11127-023-01097-2. URL <https://link.springer.com/article/10.1007/s11127-023-01097-2>.
- Office of Management and Budget. M-26-04: Increasing public trust in artificial intelligence through unbiased AI principles, 2025. URL <https://www.whitehouse.gov/wp-content/uploads/2025/12/M-26-04-Increasing-Public-Trust-in-Artificial-Intelligence-Through-Unbiased-AI-Principles.pdf>. Implementation guidance for Executive Order 14319; names additive system instructions among “Governance tools” and accepts system-prompt disclosure as neutrality-compliance evidence.
- Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/7f1f0218e45f5414c79c0679633e47bc-Abstract-Conference.html.
- Inioluwa Deborah Raji and Joy Buolamwini. Actionable auditing: Investigating the impact of publicly naming biased performance results of commercial AI products. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society (AIES ’19)*, pp. 429–435, Honolulu, HI, USA, 2019. Association for Computing Machinery. doi: 10.1145/3306618.3314244. URL <https://doi.org/10.1145/3306618.3314244>.
- Inioluwa Deborah Raji, Andrew Smart, Rebecca N. White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson, Jamila Smith-Loud, Daniel Theron, and Parker Barnes. Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT* ’20)*, pp. 33–44. Association for Computing Machinery, 2020. doi: 10.1145/3351095.3372873. URL <https://doi.org/10.1145/3351095.3372873>.
- Paul Röttger, Valentin Hofmann, Valentina Pyatkin, Musashi Hinck, Hannah Kirk, Hinrich Schuetze, and Dirk Hovy. Political compass or spinning arrow? Towards more meaningful evaluations for values and opinions in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 15295–15311, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.816. URL <https://aclanthology.org/2024.acl-long.816/>.

- David Rozado. The political preferences of LLMs. *PLOS ONE*, 19(7):e0306621, 2024. doi: 10.1371/journal.pone.0306621. URL <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0306621>.
- David Rozado and Philip Tetlock. Have we been measuring AI political bias wrong? A better approach is possible. Substack essay, June 2026. URL <https://davidrozado.substack.com/p/have-we-been-measuring-ai-political>. Published June 22, 2026. Introduces the Viewpoint Preferences–Epistemic Integrity (VPEI) framework; companion article in *Theory and Society*, doi:10.1007/s11186-026-09719-6.
- Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. Whose opinions do language models reflect? In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 29971–30004. PMLR, 2023. URL <https://proceedings.mlr.press/v202/santurkar23a.html>.
- Timo Spinde, Manuel Plank, Jan-David Krieger, Terry Ruas, Bela Gipp, and Akiko Aizawa. Neural media bias detection using distant supervision with BABE - bias annotations by experts. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pp. 1166–1177, Punta Cana, Dominican Republic, 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.findings-emnlp.101. URL <https://aclanthology.org/2021.findings-emnlp.101/>.
- Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/ed3fea9033a80feal376299fa7863f4a-Abstract-Conference.html.
- Emiel van Miltenburg, Chris van der Lee, and Emiel Krahmer. Preregistering NLP research. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 613–623, Online, 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.naacl-main.51. URL <https://aclanthology.org/2021.naacl-main.51/>.
- Supriti Vijay, Aman Priyanshu, and Ashique KhudaBukhsh. When neutral summaries are not that neutral: Quantifying political neutrality in LLM-generated news summaries (student abstract). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39(28), pp. 29514–29516, 2025. doi: 10.1609/aaai.v39i28.35308. URL <https://ojs.aaai.org/index.php/AAAI/article/view/35308>. AAAI-25 Student Abstract and Poster Program; extended version: arXiv:2410.09978.
- Xinyi Yang, Runzhe Zhan, Derek F. Wong, Shu Yang, Junchao Wu, and Lidia S. Chao. Rethinking prompt-based debiasing in large language models, 2025. URL <https://arxiv.org/abs/2503.09219>.
- Jiayi Ye, Yanbo Wang, Yue Huang, Dongping Chen, Qihui Zhang, Nuno Moniz, Tian Gao, Werner Geyer, Chao Huang, Pin-Yu Chen, Nitesh V. Chawla, and Xiangliang Zhang. Justice or prejudice? Quantifying biases in LLM-as-a-judge. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL <https://openreview.net/forum?id=3GTtZFiajM>.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhaghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023. URL <https://arxiv.org/abs/2306.05685>.